These kind of calculations are pretty simple, but the number of calculations per unit time is enormous. Which is why GPUs are RISC(Reduced Instruction Set Computer), i.e. they can handle a lot of simple calculations efficiently. CPUs on the other hand are more versatile and support a lot of instructions which makes them ideal for handling anything under the sun, however, this does mean that they take a significant hit in the time needed to perform calculations. So you now have an SoC where the CPU decides what needs to be calculated by whom and accordingly delegates the visual processing to the GPU, the audio processing to the audio processor and handles the rest of the workload by itself.

How does it help then?

While the CPU is capable of handling everything on its own in its own sweet time, the end result is a very laggy user experience. This is pretty evident in old Android devices where the delegation of graphical processing was not handled properly and the phones couldn't play high-resolution videos. But this has changed, with instruction sets very similar to desktop variants being replicated on the mobile platform even playing 1080p videos have become a mainstream function on smartphones and tablets. Here is some of the tasks that a GPU commonly handles:

- Image processing (Adding various effects like B/W, Sepia, etc to photos)
- 2D processing (Mostly used to render elements of the user interface)
- 3D processing (Games and user interface elements)
- Video decoding and encoding (This is what helps play videos on your mobile device)

We can now play heavy duty games like Doom 3 on an android phone! Don't believe us? Here is a link:

http://dgit.in/19ljm6p

This might make you wonder why even with a GPU your device still lags quite a lot. Well this depends a lot on the way the operating system implements task handling and workload distribution. Variation of hardware is also a major factor. Take the Android OS, since Android was created to run on a plethora of different CPU & GPU combinations the programmers had to ensure that it worked smoothly on phones lacking powerful GPUs as well. So quite a lot of the graphical processing normally handled by the GPU was given to the CPU, purely for the sake of maintaining compatibility. That's why an Android interface used to lag a bit even on powerful devices. But this was in the past, with Android 4.0, Google rightly took care of this issue